

NISHANT KUMAR

AI Research & Software Engineering Intern

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EDUCATION

Bharati Vidyapeeth College of Engineering, Pune

Expected June 2026

Bachelor of Technology (IT) | Current CGPA: 9.45/10

Specialization: AI/ML Focus | Mathematics Topper (Discrete Maths & Calculus).

Rashtriya Military School, Belgaum

Graduated 2023

Higher Secondary Education | Leadership Rank: School Adjutant

TECHNICAL SKILLS

- **AI/ML Frameworks:** PyTorch, TensorFlow/Keras, LangChain, FAISS Vector DB, OpenCV, SpaCy.
- **LLM Engineering:** RAG Pipelines, 4-bit Quantization (AWQ/GPTQ), Prompt Engineering.
- **Languages:** Python (Advanced), SQL, C++ (Intermediate), TypeScript.
- **Research & Tools:** Grad-CAM (Interpretability), LaTeX, Git, Linux, Docker.

AI RESEARCH & APPLIED ML PROJECTS

Hippocrates AI | AI Research Lead

Sep 2024 – Nov 2024

- Engineered a medically-aware **Retrieval-Augmented Generation (RAG)** system using Mistral-7B and FAISS, achieving an ****89.2% diagnostic accuracy**** on complex clinical cases.
- Optimized inference latency from 8s to **2.4s** by implementing 4-bit quantization (AWQ) and KV-caching for real-time diagnostic queries.
- Reduced AI hallucinations by **94%** through a citation-based verification system that anchors model responses to peer-reviewed medical literature.

Pneumonia Detection CNN | ML Engineering Lead

Aug 2024

- Developed a custom Convolutional Neural Network (CNN) trained on 5,863 chest X-rays, prioritizing a **99% Medical Recall** to eliminate false negatives in critical settings.
- Integrated **Grad-CAM (Gradient-weighted Class Activation Mapping)** to visualize AI decision-making, allowing clinicians to verify the exact lung regions identified by the model.

Hybrid Cryptography Framework | Research Contributor

Jun 2024 – Aug 2024

- Authored research (under peer review) on a hybrid **AES-256 + RC6** encryption model; achieved 1.8x faster encryption speeds compared to standard sequential implementations.
- Conducted rigorous performance benchmarking across variable data payloads to validate cryptographic robustness and computational efficiency.

SYSTEMS & LEADERSHIP

- **Arogya Saathi (SIH 2024 Finalist):** Lead architect for a healthcare platform recognized as a National Finalist (Top 2% of 100k+ teams); implemented Redis-based distributed locking for transactional integrity.
- **Military Discipline:** Developed resilient problem-solving and cross-functional teamwork in a high-pressure military school environment.
- **Technical Pitching:** Vetted by government officials and SIH judges for the strategic viability of AI-driven public sector solutions.